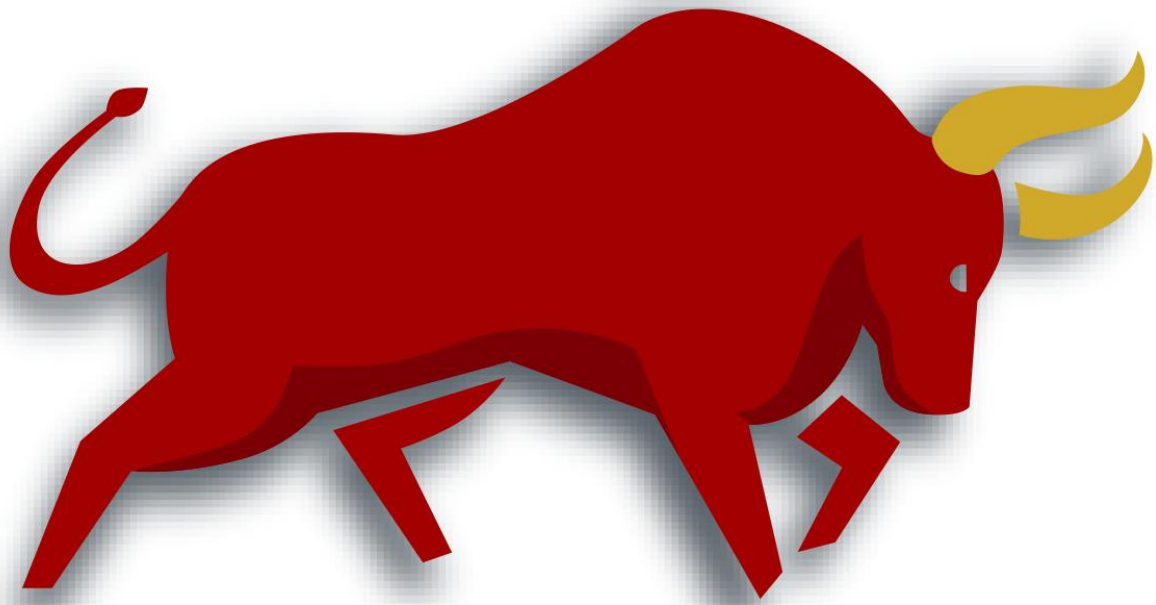

FDT Big Data Centre



FDT Scoring Platform

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Overview

This project aims to

Goals

1. Create each user a profile
2. Grade each user with a FDT score and other types of scores
3. Visualize each user's profile

System Architecture

The system architecture is shown in Figure 1:

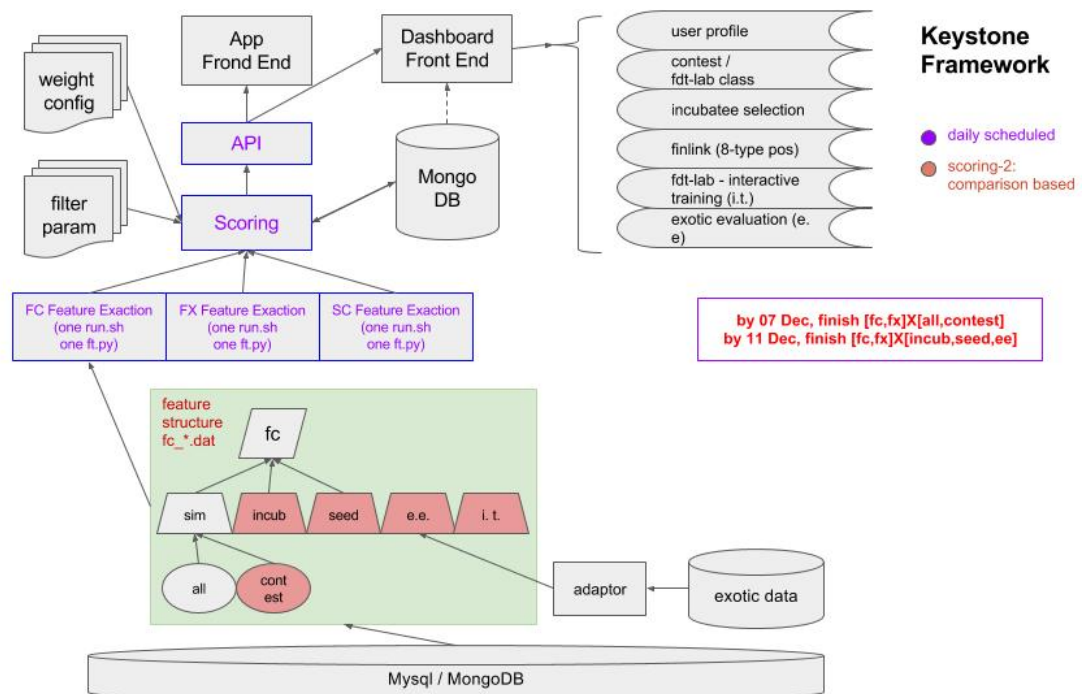


Figure 1: The Architecture of the User-Profile-Modeling Project (By Qifeng)

Note: we support different types of products: fx, fc and sc and different types of data sources. In the bottom tree structure, each branch from a product (e.g., fc) to the leaf defines a **unique data source**. (!Important!!) We maintain **exactly one copy of feature repository** (there are 19 features for now), where each feature extractor will extract features for each data source.

Challenges

The “FDT Scoring Platform” needs to handle the following challenges:



We solved these challenges by the following **Three-One** design strategies:

1. **One single feature repository** shared by all data sources and all applications. Features can be selected based on data sources and applications.
2. **One tree structure** to represent multiple data sources, where each branch uniquely defines a data source.
3. **One weight schema database** to store various weights schemas, such 8 weights files for 8 types of financial candidates in FinLink.

System Implementation

The framework for the “FDT Scoring Platform” project (risk, preference and fdt scores) has been established.

Overview

The framework for the “FDT Scoring Platform” project (risk, preference and fdt scores) has been established. It contains four major components:

1. Feature engineering scripts
2. Script execution engine
3. MongoDB for saving features
4. APIs for other applications

There are two background jobs running:

- 1) **feature-extraction**: this is the script-execution engine which runs the scripts in the feature-extractors folder (/opt/user-profile/features). This job is daily scheduled.
- 2) **compute-scores**: this is the feature-rating engine which rates each feature by an algorithm like grading students in a course. The score range is [0, 100]. (MongoDB)

Feature Engineering

We design a lot of features for calculating the FDT score. Each feature is a folder with necessary scripts and must be located in the folder “/opt/user-profile/features”. The detailed documents for current features are available at:

https://docs.google.com/a/hkfddt.com/document/d/1nyrBkxmSzY37K98gqsWibL_k3L5MUNV4P42fm0wzWh8/edit?usp=sharing

Steps to Add a New Feature

To ensure the project quality, the following code conventions for adding new features to the framework:

1. Create the main script "run.sh" which will create the file "data_source.dat" (e.g., **fx_sim_all.dat**) (Note: the first line of run.sh should be: "#!/bin/sh", which is a comment on shell location.)
2. Create a folder "dataset" which contains the files obtained from MySQL or other data sources
3. Create an optional "debug" folder if you want to write some debug files (mkdir -p debug)
4. Write a "README.txt" file to describe your feature
5. (Optional) Create a file "**test.txt**" to contain the top 10 rows of your "feature.dat" output
6. Copy your features to the folder /opt/user-profile/features/ (python and shell scripts, no generated data files or temp files, which should be "run.sh", "*.py", and "README.txt")

Note: To support multiple data sources, we revises step one as follows:

!!Important!!

Create the main script "run.sh" which will create one .dat file for each data source, **each data source should have exactly one .dat file**, such as **fx_sim_all.dat** or **fc_sim_all.dat**. **You can write multiple python scripts for fx, fc and sc or reuse the same script file for fx, fc and sc.**

!!Important!!

The code conventions can be checked automatically. An example command is like this:

```
[pengfei@iZ23bz3npd1Z user-profile-daemon]$ make check-code
```

```
Feature: features/ft_fc_avg_dr
Error: features/ft_fc_avg_dr/README.txt not found.
Info: features/ft_fc_avg_dr/test.dat found.
Info: features/ft_fc_avg_dr/run.sh found.
Info: features/ft_fc_avg_dr/run.sh outputs feature.dat.
Feature: features/ft_fc_bgl
Error: features/ft_fc_bgl/README.txt not found.
Info: features/ft_fc_bgl/test.dat found.
Info: features/ft_fc_bgl/run.sh found.
Info: features/ft_fc_bgl/run.sh outputs feature.dat.
Feature: features/ft_fc_bgw
Error: features/ft_fc_bgw/README.txt not found.
Info: features/ft_fc_bgw/test.dat found.
Info: features/ft_fc_bgw/run.sh found.
Info: features/ft_fc_bgw/run.sh outputs feature.dat.
```

Support External Data Sources

We extend the project further to support multiple external data sources. This is challenging due to various data sources, different weighting schemes, various scoring strategies. The implementation is as follows:

- 1) Create a new Mongo collection “externalUser” to save all users from external data sources;
- 2) Similar to fx, fc and sc, all external data sources are prefixed with “ex”;
- 3) Each external data source has a unique key, e.g., ex_source_1, which will be used as feature filename, weights schema filename and so on;
- 4) Specific APIs will be developed for externalUser;

MongoDB Design

We design a MongoDB named “UserProfile”, which consists of the collections of “user” and “extUser” for now.

Weights Schemes

Each application will have a specific weight scheme. All weights schemes are located in the folder “weights”.

Front End Visualization

We mainly use radar charts to visualize the scores for each user. The demo page can be found at <http://121.43.73.64/#/index/userScore>.

System Operation

Source Code

Currently, the source code is located at `/opt/user-profile-daemon` on the aliyun server (10.139.96.26).

Folder	Comments
<code>/opt/user-profile-daemon/</code>	Source code
<code>/var/log/user-profile-daemon/</code>	Log files
<code>/opt/user-profile/features/</code>	Feature files

Running Project

When you are in the folder of `/opt/user-profile-daemon/`, you can easily run the project using the commands below:

- (1) Compile the project again only if you changed the source code

```
[pengfei@iZ23bz3npd1Z user-profile-daemon]$ make compile-project
```

(2) Run the project

```
[pengfei@iZ23bz3npd1Z user-profile-daemon]$ make run-project
```

Please make sure there is only one process running for user-profile-daemon. Kill the old process if you want to run a new user-profile-daemon, as shown in the below figure by the “top” command. The daemon is a Java process named “user-profile-daemon-0.0.1-SNAPSHOT.jar”

```
top - 14:42:33 up 42 days, 21:11, 11 users, load average: 0.15, 0.07, 0.41
Tasks: 222 total, 1 running, 221 sleeping, 0 stopped, 0 zombie
Cpu(s):  0.0%us,  1.2%sy,  0.0%ni, 98.8%id,  0.0%wa,  0.0%hi,  0.0%si,  0.0%st
Mem:   65971132k total, 35595904k used, 30375228k free,  433560k buffers
Swap:      0k total,      0k used,      0k free, 32014004k cached
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
26505	pengfei	20	0	99040	72m	1144	S	9.8	0.1	180:54.89	tmux -2 -f /usr/share/byobu/profiles/tmuxrc new-session -n - /usr/bin/byobu-shell
28371	pengfei	20	0	15160	1336	936	R	9.8	0.0	0:00.29	top
2212	pengfei	20	0	105m	1916	1432	S	0.0	0.0	0:00.09	/bin/bash
3497	pengfei	20	0	105m	1992	1436	S	0.0	0.0	0:01.26	/bin/bash
4344	pengfei	20	0	105m	1824	1408	S	0.0	0.0	0:00.00	/bin/bash
6408	pengfei	20	0	105m	1856	1432	S	0.0	0.0	0:00.09	/bin/bash
8601	pengfei	20	0	105m	1860	1432	S	0.0	0.0	0:00.04	/bin/bash
8778	pengfei	20	0	98.3m	1944	908	S	0.0	0.0	0:02.80	sshd: pengfei@pts/5
8779	pengfei	20	0	105m	1792	1412	S	0.0	0.0	0:00.00	-bash
8798	pengfei	20	0	103m	1424	1124	S	0.0	0.0	0:00.01	/bin/sh -e /usr/bin/byobu-select-session
8831	pengfei	20	0	23712	1124	816	S	0.0	0.0	0:00.00	attach -t 1
9251	pengfei	20	0	105m	1924	1452	S	0.0	0.0	0:00.41	/bin/bash
12137	pengfei	20	0	98.7m	948	728	S	0.0	0.0	0:00.00	make run-project
12138	pengfei	20	0	103m	1160	996	S	0.0	0.0	0:00.00	/bin/sh -c mvn clean package; java -jar target/user-profile-daemon-0.0.1-SNAPSHOT.jar
12231	pengfei	20	0	98.1m	1984	988	S	0.0	0.0	0:00.46	sshd: pengfei@notty
12232	pengfei	20	0	57240	2272	1664	S	0.0	0.0	0:00.00	/usr/libexec/openssh/sftp-server
12235	pengfei	20	0	21.3g	1.0g	12m	S	0.0	1.6	1:24.82	java -jar target/user-profile-daemon-0.0.1-SNAPSHOT.jar
13252	pengfei	20	0	105m	1840	1432	S	0.0	0.0	0:00.01	/bin/bash
15410	pengfei	20	0	105m	1904	1436	S	0.0	0.0	0:00.31	/bin/bash
16407	pengfei	20	0	60188	3736	2888	S	0.0	0.0	0:00.23	ssh mongodb
26507	pengfei	20	0	105m	1948	1448	S	0.0	0.0	0:00.15	/bin/bash
28249	pengfei	20	0	105m	1868	1448	S	0.0	0.0	0:00.38	/bin/bash
31202	pengfei	20	0	105m	1880	1448	S	0.0	0.0	0:00.29	/bin/bash
31949	pengfei	20	0	105m	2048	1456	S	0.0	0.0	0:01.77	/bin/bash

Scheduling Jobs on-demand

The platform now supports scheduling the jobs on demand:

(1) Invoke all run.sh

```
curl http://10.139.96.26:8099/api/scores?job=compute
```

This job may take around **10 minutes** to finish.

(2) Save features to mongodb

```
curl http://10.139.96.26:8099/api/scores?job=save
```

You can run these commands using your own account on aliyun.

Tuning Weights

The weights of features need to be tuned from time to time. We maintain a separate weights file for the project, which can be changed on demand.

Compared with the old properties files based solution, we have developed a new XML based version, with the following advantages:

- 1) Weights files are now stored in more structured XML files;
- 2) Two weights files are provided: trader.xml (effective by default) and banker.xml.
- 3) The effective weights file can be changed in application.properties, e.g., to be banker.xml.
- 4) A new group of features can be added directly to the XML file, e.g., trader.xml. (Please double check the feature names to be the same as /opt/user-profile/features).

Note that, the project needs to be restarted for any changes to the configurations files.

We currently support both “banker.xml” (for banker scenario) and “trader.xml” (for trader scenario). An example command is like this:

```
[pengfei@iZ23bz3npd1Z user-profile-daemon]$ vim src/main/resources/banker.xml
```

Or

```
[pengfei@iZ23bz3npd1Z user-profile-daemon]$ vim src/main/resources/trader.xml
```

We need to specify which file to use in “application.properties”.

```
[pengfei@iZ23bz3npd1Z user-profile-daemon]$ vim src/main/resources/application.properties
```

```
server.port = 8098
```

```
weights.file = trader.xml
```

```
<?xml version="1.0" encoding="UTF-8" ?>
<features>
  <consistency>

<ft_fc_std_op_dr>0.5</ft_fc_std_op_dr>

<ft_fc_std_op_qty>0.5</ft_fc_std_op_qty>
  </consistency>
  <activity>
    <ft_login_cnt>0.4</ft_login_cnt>
    <ft_trade_cnt>0.6</ft_trade_cnt>
  </activity>
  <profitability>
    <ft_fc_bgw>0.2</ft_fc_bgw>
    <ft_up>0.4</ft_up>
    <ft_streak>0.4</ft_streak>
  </profitability>
  <riskCtrl>

<ft_avg_ngt_op>0.15</ft_avg_ngt_op>

<ft_avg_max_op>0.15</ft_avg_max_op>
```

Any group of weights can be changed, but please make sure the feature names are correct and **restart** the project after changing the weights or updating the application.properties file.

Support FinLink

We extend the framework to support FinLink. We have developed a preliminary demo to select 8 types of financial candidates of China Merchants Bank (CMB).

Type	Characteristics	Weight File	Comment
banker_trader_steady	short op duration & good risk control	banker_trader_steady.xml	稳健交易型
banker_trader_active	short op duration & good profitability	banker_trader_active.xml	积极交易型
banker_trader_balanced	short op duration & good balance	banker_trader_balanced.xml	平衡交易型
banker_investor_steady	long op duration & good risk control	banker_investor_steady.xml	稳健投资型
banker_investor_active	long op duration & good profitability	banker_investor_active.xml	积极投资型
banker_investor_balanced	long op duration & good balance	banker_investor_balanced.xml	平衡投资型
banker_risk_control	low risk preference good profitability	banker_risk_control.xml	风险控制型
banker_marketing	good activity good usage	banker_marketing.xml	市场营销型